SOLOMON GOLDGRABER CASSPI

Algorithm Engineer

052-5903757 | solomona89@gmail.com
Jerusalem, Israel | Solomon Goldgraber-Casspi

ACADEMIC PROJECTS

Multi-Agent Control

- **Consensus control protocols**
- **Formation control via rigidity theory on graph network systems**
- **Analysis and design of network dynamical systems**

DVS Camera

- **Neuromorphic cameras-based drone detection.**

Spatial SP

- **DOA estimation: Bartlett (DAS), Null BF, MVDR, MUSIC, Poly. Rooting.**
- **Eigen structure-based pre-processing techniques: Spatial-Smoothing, Forward-Backward.**
- **Performance Analysis Of Signal Detection in The Presence of Gain and Phase Imbalance Using Eigen Structure-Based Pre-Processing Methods and Information-Theoretic Criteria.**

Mapping & Perception for Autonomous Robot

- **Implementation of Probabilistic Occupancy Map**
- **2D pose estimation with Kalman filter, Extended Kalman filter, EKF-SLAM, FastSLAM 1.0, Particle Filter**
- **Vision - Monocular Visual Odometry, DNN - Segmentation (via maskRCNN), Object Detection (via Faster-RCNN, YOLOv5)**

OFDM MIMO

- **Performance Analysis of (SISO, MRC, STC, EigenBF, SM) Schemes.**
- **OFDM Simulation.**

Computer Vision

- **Homography & Panorama.**
- **Object Detection, Classification, and Localization Based YOLOv3.**

EDUCATION

2023 - 2024 **PhD Candidate (Ph.D.) in EE & Computer Engineering** — **Ben-Gurion University**
S. G. Casspi, O. Hüsser, G. Revach and N. Shlezinger, "LQGNET: Hybrid Model-Based and Data-Driven Linear Quadratic Stochastic Control," ICASSP 2023 - IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP), Rhodes Island, Greece.

2018 - 2022 **Master of Science (M.Sc.) in EE Engineering, WAM: 89** — **Tel Aviv University**
S. G. Casspi, J. Tabrikian and H. Messer, "Blind Array Calibration of Mutual Coupling, Phase, and Gain, for Automotive Radar," in IEEE Transactions on Aerospace and Electronic Systems (patented).

2012 - 2016 **Bachelor of Science (B.Sc.) in EE Engineering, WAM: 88.7** — **Holon Institute of Technology**
Characterization And Monitoring of Transients in Smart Grid Signals With Hilbert-Huang Transform - Dr. Yuval Beck Lab.

EXPERIENCE

2023 – 2024 **TA - Introduction to Numerical Analysis and Computation** — **Ben-Gurion University**

2019 – 2023 **Radar for Automotive PHY Algorithm (Part-Time)** — **Intel corp., Mobileye**

- Offline - Tx DC calibration algorithm.
- Fixed-point design of correlator module.
- Developing Rx Analyzer with MATLAB.
- Modeling real-world automotive scenes and testing for Radar system with HFSS.
- Extensive test performance environment developing with MATLAB.

2018 – 2019 **DSP & PHY engineer (Full-Time)** — **Tracespan Communications**

- Development and implementation algorithms of PHY (L1) LTE communication analyzer.
- Development and implementation algorithms of PHY (L1) G.fast communication analyzer.

2016 – 2018 **Integration & Test Engineer (Full-Time)** — **Qualcomm**

- LTE communication protocol background.
- Test developer for physical layer (PHY).

LANGUAGES

Hebrew - Native, **English** - Fluent,
Russian - Conversational

SKILLS

Programming:MATLAB, Python, C++, C.